

CYCLIC EUCLIDEAN-QUOTIENT QUEUES: TERMINAL CORES, DEPTH LAYERS, AND SMALL FIBRES

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ABSTRACT. Fix the sum N of the canonical continued-fraction digits of a rational in $(0, 1)$. We move its first Euclidean quotient to the back of the queue and, when necessary, apply the terminal-one identity that restores the unique finite expansion. The entrance time is exactly the position of the last quotient equal to one, with sharp maximum $N - 2$. We give a cyclic run-absorption decoder for the terminal core and identify the eventual period with its primitive rotation period. Formal generating functions enumerate every exact-depth layer. Two explicit inverse branches give every one-step fibre, the image, and all Garden states; fibres have size at most two. A second derivation uses the alternating run blocks of the subtractive Euclidean path. Restricted compositions, Stern–Brocot coding, continuants, and necklace enumeration are treated as background. The owner search is bounded; no novelty, priority, or external-release claim is made.

1. AN HONEST RATIONAL LEVEL AND THE QUEUE MAP

Every rational $q \in (0, 1)$ has a unique canonical finite regular continued fraction

$$q = [0; a_1, \dots, a_k], \quad a_i \geq 1, \quad a_k \geq 2.$$

For $N \geq 2$, define

$$\mathcal{R}_N = \{q \in \mathbb{Q} : 0 < q < 1, \sum_i a_i = N\}.$$

The levels $N = 0, 1$ are empty. Subtracting one from the final digit identifies the digit words in $\mathcal{R}_N$ with the positive compositions of $N - 1$, so

$$(1) \quad |\mathcal{R}_N| = 2^{N-2}.$$

To fix the Euclidean convention, write $q = u/v$ in lowest terms. Repeatedly subtract the smaller positive coordinate from the larger, recording L for $v \leftarrow v - u$ and R for $u \leftarrow u - v$. At equality, repeat the preceding letter and send that coordinate to zero. The run word is

$$(2) \quad E(q) = L^{a_1} R^{a_2} L^{a_3} \dots,$$

with alternating letters, and its length is N . This follows by grouping successive subtractions between exchanges of the smaller coordinate; the last equality step belongs to the final quotient, explaining $a_k \geq 2$. Thus N is the step count for this explicitly stated subtractive algorithm. Euclidean cost statistics are established territory [5], and the Stern–Brocot/run-word relation is classical [6]; both receive zero contribution credit.

On digit words put

$$(3) \quad \Phi(a_1, \dots, a_k) = \begin{cases} (a_1), & k = 1, \\ (a_2, \dots, a_k, a_1), & k > 1, \ a_1 > 1, \\ (a_2, \dots, a_{k-1}, a_k + 1), & k > 1, \ a_1 = 1. \end{cases}$$

The final branch is rotation followed by the canonical identity $[\dots, b, 1] = [\dots, b + 1]$. It is part of the definition, not an optional postprocessing step. Digit sum and the terminal condition are preserved, so Φ is a self-map of every $\mathcal{R}_N$.

Finite words, continuants, prefixes, endings, and cyclic permutations are well-developed interfaces [3, 4]. Cyclic and restricted-part compositions likewise have direct enumerative owners [1, 2]. All uniqueness, continuant, composition, regular-language, Burnside, and divisor-counting machinery is zero credit. Internally, earlier systems already occupy cyclic run reduction, sharp-linear-depth/fibre packages, and synchronous composition refinement. The last one has logarithmic absorption to one fixed point and an all-iterate suffix kernel; the present map is a one-place quotient queue with linear depth, fibres at most

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two, and many rotation cycles. This proves nonconjugacy, not priority. The only residual claim is the exact temporal conjunction for (3).

2. THE MARKER CLOCK AND TERMINAL CORE

For a digit word $w = (a_1, \dots, a_k)$, set

$$\delta(w) = \max\{i : a_i = 1\},$$

with $\delta(w) = 0$ when no digit equals one. View the indices cyclically. For every maximal run of ones, add its length to the non-one digit immediately preceding it and delete the run. Finally cut the resulting cyclic word immediately after the last original one; call the ordered word $\kappa(w)$. If there was no one, put $\kappa(w) = w$.

Theorem 2.1 (terminal-core theorem). *For every $w \in \mathcal{R}_N$,*

$$(4) \quad \text{depth}(w) = \delta(w), \quad \Phi^{\delta(w)}(w) = \kappa(w).$$

The recurrent set is exactly the words whose digits are all at least two. On it Φ is left rotation. Hence the eventual period of w is the primitive rotation period of $\kappa(w)$. The maximum depth on $\mathcal{R}_N$ is $N - 2$, attained by $(1^{N-2}, 2)$.

Proof. If the leading digit exceeds one, (3) moves it behind every one; if it equals one, the update deletes it and increments the current last digit, which is at least two. No step creates a one. Therefore the largest index occupied by a one decreases by exactly one at each step until it vanishes, proving the first formula.

Track a maximal cyclic run of s ones. When its leading member reaches the front, deletion contributes one to the digit preceding that run in cyclic order; repeating this s times adds s there. Runs do not interfere because no new one is created. After the last original one is deleted, the next original non-one digit is at the front, which is exactly the prescribed cut. This proves (4). A word with no one is merely rotated, so it is recurrent and has the stated primitive period; every other word has positive depth. Finally, a word of weight N ending at least two has at most $N - 2$ digits before its end, and the displayed witness realizes that bound. $\square$

2.1. Independent Euclidean-path route. Mark the maximal constant blocks in the subtractive word (2). The quotient queue removes the first block, appends a block of the same length, and reassigns alternating L/R labels beginning with L . If the appended block has length one, it consists only of the terminal equality step. The canonical terminal convention absorbs that step into the preceding block, changing lengths $(\dots, b, 1)$ to $(\dots, b + 1)$. Thus this block operation reconstructs (3) directly from the rational pair algorithm.

In this coordinate, digits equal to one are exactly singleton Euclidean blocks. Each queue step shifts every marked singleton one block toward the front, and the front singleton is suppressed at the terminal end. The last singleton therefore survives for precisely its block index, while successive suppression adds each singleton run to its cyclic predecessor. This recovers both assertions in (4) without using the digit recurrence proof. It also gives the two inverse path operations used in theorem 4.1: move an admissible final block to the front, or split one terminal equality step from a final block of length at least three and insert it as a new singleton first block.

3. EVERY EXACT-DEPTH LAYER

Let $D_t(x)$ be the formal ordinary generating function in digit sum for states of exact depth t . A recurrent word is a nonempty sequence of parts at least two. If $t \geq 1$, its first $t - 1$ parts are arbitrary positive parts, part t equals one, and its nonempty suffix has all parts at least two. Consequently

$$(5) \quad D_0(x) = \frac{x^2}{1 - x - x^2}, \quad D_t(x) = \frac{x^{t+2}}{(1 - x)^{t-1}(1 - x - x^2)} \quad (t \geq 1).$$

With $F_0 = 0, F_1 = 1$, this says

$$[x^N]D_0 = F_{N-1}, \quad [x^N]D_1 = F_{N-2},$$

and, for $t \geq 2$ and $M = N - t - 2 \geq 0$,

$$(6) \quad [x^N]D_t = \sum_{j=0}^M \binom{j+t-2}{t-2} F_{M-j+1};$$

the coefficient is zero when $N < t + 2$. As a boundary check,

$$(7) \quad \sum_{t \geq 0} D_t(x) = \frac{x^2}{1 - 2x},$$

whose weight- N coefficient is (1).

Proof. The atomic series for a positive part is $x/(1-x)$ and for a part at least two is $x^2/(1-x)$. The two sequence decompositions above give (5). Expanding $(1-x)^{-(t-1)}$ and $(1-x-x^2)^{-1} = \sum_{m \geq 0} F_{m+1}x^m$ gives (6). Summing the geometric series over t yields (7). $\square$

4. EVERY ONE-STEP FIBRE

For a target $y = (b_1, \dots, b_\ell)$, define the possible rotation inverse

$$\rho(y) = \begin{cases} y, & \ell = 1, \\ (b_\ell, b_1, \dots, b_{\ell-1}), & \ell > 1, b_{\ell-1} \geq 2, \end{cases}$$

when the displayed branch applies, and the possible deletion inverse

$$\eta(y) = (1, b_1, \dots, b_{\ell-1}, b_\ell - 1) \quad \text{when } b_\ell \geq 3.$$

Theorem 4.1 (complete fibre law). *The fibre of y consists exactly of the applicable, distinct words $\rho(y)$ and $\eta(y)$. Hence every fibre has size 0, 1, or 2. Moreover*

$$(8) \quad |\text{im } \Phi| = \begin{cases} 1, & N = 2, 3, \\ 3 \cdot 2^{N-4}, & N \geq 4, \end{cases}$$

and the number of Garden states is 0 for $N = 2$, 1 for $N = 3$, and 2^{N-4} for $N \geq 4$.

Proof. The first two branches of (3) invert only by moving the last digit to the front; for length greater than one the resulting last digit is $b_{\ell-1}$, so canonicity is exactly the displayed condition. The last branch can invert only by inserting a leading one and removing one from the last digit, which is legal exactly when $b_\ell \geq 3$. These sources have different lengths and direct substitution verifies them, so there are no others and no collision.

A target has no inverse precisely when it has length at least two and ends in (1, 2). For $N = 3$ this is the sole word (1, 2). For $N \geq 4$, deleting that suffix leaves an arbitrary positive composition of $N - 3$, giving 2^{N-4} Garden states. Subtracting from (1) proves (8); $N = 2$ is immediate. $\square$

5. RECURRENT NECKLACES

For recurrent words of weight N and length k , Burnside's lemma gives the number of rotation orbits

$$(9) \quad \mathcal{C}_{N,k} = \frac{1}{k} \sum_{d \mid \gcd(N,k)} \varphi(d) \binom{N/d - k/d - 1}{k/d - 1}, \quad 1 \leq k \leq \lfloor N/2 \rfloor.$$

Indeed a word fixed by a rotation with repetition factor d reduces to a length- k/d composition of N/d with parts at least two. The binomial is well-defined because $N \geq 2k$. Formula (9) counts cycles by word length, not by primitive dynamical period; no zeta formula is inferred. Fixed points are the constant digit words, so

$$(10) \quad |\text{Fix } \Phi| = d(N) - 1,$$

where $d(N)$ is the number of positive divisors. Both formulas are classical necklace consequences and receive zero contribution credit; their role is only to complete the recurrent picture produced by [theorem 2.1](#).

6. EXACT CONTROL AND LIMITATIONS

The deterministic verifier constructs the rational numerator/denominator by continuants, independently reconstructs its subtractive L/R path, and exhausts every state for $2 \leq N \leq 18$. It checks the path blocks, terminal core, exact depth, primitive period, every target fibre, all layer counts, image and Garden counts, Burnside orbits, and fixed points. These finite checks are falsification evidence, not proofs.

The map is specific to canonical finite expansions in $(0, 1)$ and to the stated terminal convention. It is not a map on all positive rationals, not a Gauss/Farey shift, and not an assertion about general continued fractions or continuants. The owner screen was bounded and cannot certify absence. Novelty, priority, authorship, posting, submission, and every external-release decision remain on hold.

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